chapter

21

Lipid Biosynthesis

1. **Pathway of Carbon in Fatty Acid Synthesis** Using your knowledge of fatty acid biosynthesis, provide an explanation for the following experimental observations:
   (a) Addition of uniformly labeled [^{14}C]acetyl-CoA to a soluble liver fraction yields palmitate uniformly labeled with ^{14}C.
   (b) However, addition of a *trace* of uniformly labeled [^{14}C]acetyl-CoA in the presence of an excess of unlabeled malonyl-CoA to a soluble liver fraction yields palmitate labeled with ^{14}C only in C-15 and C-16.

   Answer Recall that "loading" of the acyl carrier protein requires an initial addition of acetyl-CoA, followed by the addition of malonyl-CoA. Malonyl-CoA is normally produced by the addition of CO_2 to acetyl-CoA by acetyl-CoA carboxylase.
   (a) In the presence of an excess of [^{14}C]acetyl-CoA, the metabolic pool of malonyl-CoA becomes labeled at C-1 and C-2. This results in the formation of uniformly labeled palmitate.
   (b) If a *trace* of [^{14}C]acetyl-CoA is introduced in the presence of a large excess of unlabeled malonyl-CoA, the metabolic pool of malonyl-CoA does not become labeled. The trace of [^{14}C]acetyl-CoA is loaded onto the acyl carrier protein (to become C-15 and C-16 of palmitate) rather than transformed into malonyl-CoA (a slow, rate-controlling process). In addition, any labeled malonyl-CoA is diluted by the presence of excess unlabeled malonyl-CoA.

2. **Synthesis of Fatty Acids from Glucose** After a person has ingested large amounts of sucrose, the glucose and fructose that exceed caloric requirements are transformed to fatty acids for triacylglycerol synthesis. This fatty acid synthesis consumes acetyl-CoA, ATP, and NADPH. How are these substances produced from glucose?

   Answer Glucose and fructose are degraded to pyruvate via glycolysis in the cytosol. Pyruvate enters the mitochondrion and is oxidatively decarboxylated to acetyl-CoA, some of which enters the citric acid cycle. The reducing equivalents (NADH, $FADH_2$) produced in the citric acid cycle are used for ATP production by oxidative phosphorylation. The remainder of the acetyl-CoA is exported to the cytosol (via the acetyl group shuttle) for fatty acid synthesis (see Fig. 21–10). Some NADPH is produced in the pentose phosphate pathway (Fig. 14–21) in the cytosol, using glucose as a substrate, and some is produced by the action of malic enzyme on cytoplasmic malate.

3. **Net Equation of Fatty Acid Synthesis** Write the net equation for the biosynthesis of palmitate in rat liver, starting from mitochondrial acetyl-CoA and cytosolic NADPH, ATP, and CO_2.

   Answer Most of the acetyl-CoA used in fatty acid synthesis is formed from the oxidation of pyruvate in mitochondria. The inner mitochondrial membrane is impermeable to acetyl-CoA, and transfer across the membrane occurs via the acetyl group shuttle (see Fig. 21–10). This process requires the input of 1 ATP per acetyl-CoA

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$$\text{Acetyl-CoA}_{(\text{mit})} + \text{ATP} + \text{H}_2\text{O} \longrightarrow \text{acetyl-CoA}_{(\text{cyt})} + \text{ADP} + \text{P}_\text{i} + \text{H}^+$$

or, for palmitate,

$$8\ \text{Acetyl-CoA}_{(\text{mit})} + 8\text{ATP} + 8\text{H}_2\text{O} \longrightarrow 8\ \text{acetyl-CoA}_{(\text{cyt})} + 8\text{ADP} + 8\text{P}_\text{i} + 8\text{H}^+$$

Cytosolic acetyl-CoA ("the primer") is condensed with malonyl-CoA, then reduced, hydrated, and reduced again. This process is repeated seven times, each time adding an additional acetyl unit to the fatty acyl–CoA. The equation for this process is

$$8\ \text{Acetyl-CoA} + 14\text{NADPH} + 6\text{H}^+ + 7\text{ATP} + \text{H}_2\text{O} \longrightarrow \text{palmitate} + 8\text{CoA} + 14\text{NADP}^+ + 7\text{ADP} + 7\text{P}_\text{i}$$

(Note that this differs from Eqn 21–3 in the text because the 7 H_2O for hydrolysis of 7 ATP were omitted from Eqn 21–1, as were 7 H^+ in the products of that reaction.)

The net equation for the overall process in the cytosol is

$$8\ \text{Acetyl-CoA}_{(\text{mit})} + 15\text{ATP} + 14\text{NADPH} + 9\text{H}_2\text{O} \longrightarrow \text{palmitate} + 8\text{CoA} + 15\text{ADP} + 15\text{P}_\text{i} + 14\text{NADP}^+ + 2\text{H}^+$$

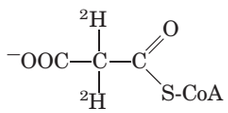
This may be viewed as the sum of two reactions:

$$8\ \text{Acetyl-CoA} + 14\text{NADPH} + 13\text{H}^+ \longrightarrow \text{CH}_3(\text{CH}_2)_{14}\text{COO}^- + 8\text{CoA} + 14\text{NADP}^+ + 6\text{H}_2\text{O}$$

$$15\text{ATP} + 15\text{H}_2\text{O} \longrightarrow 15\text{ADP} + 15\text{P}_\text{i} + 15\text{H}^+$$

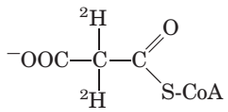
4. Pathway of Hydrogen in Fatty Acid Synthesis Consider a preparation that contains all the enzymes and cofactors necessary for fatty acid biosynthesis from added acetyl-CoA and malonyl-CoA.

(a) If [2-^{2}H]acetyl-CoA (labeled with deuterium, the heavy isotope of hydrogen)



and an excess of unlabeled malonyl-CoA are added as substrates, how many deuterium atoms are incorporated into every molecule of palmitate? What are their locations? Explain.

(b) If unlabeled acetyl-CoA and [2-^{2}H]malonyl-CoA



are added as substrates, how many deuterium atoms are incorporated into every molecule of palmitate? What are their locations? Explain.

Answer

(a) Three deuteriums per palmitate; all located on C-16; all other two-carbon units are derived from unlabeled malonyl-CoA. (Note that the unlabeled malonyl-CoA is in excess.)

(b) Seven deuteriums per palmitate; one on each even-numbered carbon except C-16. One deuterium is lost from each labeled even-numbered carbon at the dehydration step (see Fig. 21–6).

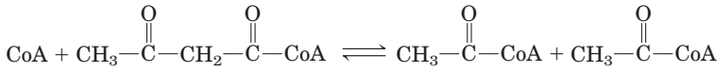
5. Energetics of β-Ketoacyl-ACP Synthase In the condensation reaction catalyzed by β-ketoacyl-ACP synthase (see Fig. 21–6), a four-carbon unit is synthesized by the combination of a two-carbon unit and a three-carbon unit, with the release of CO_2. What is the thermodynamic advantage of this process over one that simply combines two two-carbon units?

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Answer If a four-carbon unit were synthesized by combining two two-carbon units, the reaction would be reversible—for example, the reaction catalyzed by thiolase in the oxidation of fatty acids (see Fig. 17–8).



By using the three-carbon unit malonyl-CoA, the activated form of acetyl-CoA (recall that malonyl-CoA synthesis requires the input of energy from ATP), metabolite flow is driven in the direction of fatty acid synthesis by the exergonic release of CO_2.

6. Modulation of Acetyl-CoA Carboxylase Acetyl-CoA carboxylase is the principal regulation point in the biosynthesis of fatty acids. Some of the properties of the enzyme are described below.

(a) Addition of citrate or isocitrate raises the V_{max} of the enzyme as much as 10-fold.

(b) The enzyme exists in two interconvertible forms that differ markedly in their activities:

$$\text{Protomer (inactive)} \rightleftharpoons \text{filamentous polymer (active)}$$

Citrate and isocitrate bind preferentially to the filamentous form, and palmitoyl-CoA binds preferentially to the protomer.

Explain how these properties are consistent with the regulatory role of acetyl-CoA carboxylase in the biosynthesis of fatty acids.

Answer The rate-limiting step in the biosynthesis of fatty acids is the carboxylation of acetyl-CoA, catalyzed by acetyl-CoA carboxylase. High levels of citrate and isocitrate indicate that conditions are favorable for fatty acid synthesis: an active citric acid cycle is providing a plentiful supply of ATP, reducing equivalents, and acetyl-CoA. Citrate or isocitrate stimulates (increases the V_{max} of) the rate-limiting enzymatic step in fatty acid biosynthesis **(a).** Furthermore, because citrate and isocitrate bind more tightly to the filamentous form of the enzyme (the active form), the presence of citrate or isocitrate drives the protomer $\rightleftharpoons$ polymer equilibrium in the direction of the active (polymer) form **(b).** In contrast, palmitoyl-CoA (the end product of fatty acid biosynthesis) drives the equilibrium in the direction of the inactive (protomer) form. Hence, when the end product of fatty acid biosynthesis builds up, fatty acid biosynthesis is slowed down.

7. Shuttling of Acetyl Groups across the Mitochondrial Inner Membrane The acetyl group of acetyl-CoA, produced by the oxidative decarboxylation of pyruvate in the mitochondrion, is transferred to the cytosol by the acetyl group shuttle outlined in Figure 21–10.

(a) Write the overall equation for the transfer of one acetyl group from the mitochondrion to the cytosol.

(b) What is the cost of this process in ATPs per acetyl group?

(c) In Chapter 17 we encountered an acyl group shuttle in the transfer of fatty acyl–CoA from the cytosol to the mitochondrion in preparation for β oxidation (see Fig. 17–6). One result of that shuttle was separation of the mitochondrial and cytosolic pools of CoA. Does the acetyl group shuttle also accomplish this? Explain.

Answer

(a) The reactions involved in the transfer of one acetyl group are

$$\text{Acetyl-CoA}_{(mit)} + \text{oxaloacetate}_{(mit)} \longrightarrow \text{citrate}_{(mit)} + \text{CoA}_{(mit)}$$
$$\text{Citrate}_{(mit)} \longrightarrow \text{citrate}_{(cyt)}$$
$$\text{Citrate}_{(cyt)} + \text{ATP} + \text{CoA}_{(cyt)} \longrightarrow \text{oxaloacetate}_{(cyt)} + \text{ADP} + P_i + \text{acetyl-CoA}_{(cyt)}$$
$$\text{Oxaloacetate}_{(cyt)} + \text{NADH} + H^+ \longrightarrow \text{malate}_{(cyt)} + \text{NAD}^+$$
$$\text{Malate}_{(cyt)} \longrightarrow \text{malate}_{(mit)}$$
$$\text{Malate}_{(mit)} + \text{NAD}^+ \longrightarrow \text{oxaloacetate}_{(mit)} + \text{NADH} + H^+$$

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The overall equation is

$$\text{Acetyl-CoA}_{(\text{mit})} + \text{ATP} + \text{CoA}_{(\text{cyt})} \longrightarrow \text{acetyl-CoA}_{(\text{cyt})} + \text{ADP} + P_i + \text{CoA}_{(\text{mit})}$$

(b) The transfer of one acetyl group from the mitochondrial matrix to the cytosol (one turn of the acetyl group shuttle) is accompanied by the hydrolysis of ATP.

(c) The transfer requires the input of one cytosolic CoA, and one mitochondrial CoA is released. Thus, the shuttle keeps the cytosolic and mitochondrial pools of CoA separate.

8. Oxygen Requirement for Desaturases The biosynthesis of palmitoleate (see Fig. 21–12), a common unsaturated fatty acid with a cis double bond in the Δ^9 position, uses palmitate as a precursor. Can this be carried out under strictly anaerobic conditions? Explain.

Answer No; oxygen is required. The double bond in palmitoleic acid is introduced by an oxidation catalyzed by fatty acyl–CoA oxygenase (or fatty acyl–CoA desaturase; see Fig. 21–14), a mixed-function oxidase that requires molecular oxygen as a cosubstrate.

9. Energy Cost of Triacylglycerol Synthesis Use a net equation for the biosynthesis of tripalmitoylglycerol (tripalmitin) from glycerol and palmitate to show how many ATPs are required per molecule of tripalmitin formed.

Answer The reactions involved in conversion of glycerol and palmitate to tripalmitin are

$$\text{ATP} + \text{glycerol} \longrightarrow \text{glycerol 3-phosphate} + \text{ADP} + H^+$$
$$3\ \text{Palmitate} + 3\text{ATP} + 3\text{CoA} \longrightarrow 3\ \text{palmitoyl-CoA} + 3\text{AMP} + 3PP_i + 3H^+$$
$$3H_2O + 3PP_i \longrightarrow 6P_i + 3H^+$$
$$2\ \text{Palmitoyl-CoA} + \text{glycerol 3-phosphate} \longrightarrow \text{dipalmitoylglycerol 3-phosphate} + \text{CoA}$$
$$\text{Dipalmitoylglycerol 3-phosphate} + H_2O \longrightarrow \text{dipalmitoylglycerol} + P_i$$
$$\text{Dipalmitoylglycerol} + \text{palmitoyl-CoA} \longrightarrow \text{tripalmitin} + \text{CoA}$$

The overall reaction is

$$3\ \text{Palmitate} + \text{glycerol} + 4\text{ATP} + 4H_2O \longrightarrow \text{tripalmitin} + 3\text{AMP} + \text{ADP} + 7P_i + 7H^+$$

Including $3\text{AMP} + 3\text{ATP} \rightarrow 6\text{ADP}$, the overall reaction becomes

$$3\ \text{Palmitate} + \text{glycerol} + 7\text{ATP} + 4H_2O \longrightarrow \text{tripalmitin} + 7\text{ADP} + 7P_i + 7H^+$$

10. Turnover of Triacylglycerols in Adipose Tissue When [^{14}C]glucose is added to the balanced diet of adult rats, there is no increase in the total amount of stored triacylglycerols, but the triacylglycerols become labeled with ^{14}C. Explain.

Answer In adult rats, stored triacylglycerols are maintained at a steady-state level through a balance of the rates of degradation and biosynthesis. Hence, the depots of fats are continually turned over, explaining the incorporation of ^{14}C from dietary [^{14}C]glucose.

11. Energy Cost of Phosphatidylcholine Synthesis Write the sequence of steps and the net reaction for the biosynthesis of phosphatidylcholine by the salvage pathway from oleate, palmitate, dihydroxyacetone phosphate, and choline. Starting from these precursors, what is the cost (in number of ATPs) of the synthesis of phosphatidylcholine by the salvage pathway?

Answer The sequence of steps required to carry out phosphatidylcholine biosynthesis is

$$\text{Dihydroxyacetone phosphate} + \text{NADH} + H^+ \longrightarrow \text{glycerol 3-phosphate} + \text{NAD}^+$$
$$\text{Palmitate} + \text{ATP} + \text{CoA} \longrightarrow \text{palmitoyl-CoA} + \text{AMP} + PP_i$$
$$\text{Oleic acid} + \text{ATP} + \text{CoA} \longrightarrow \text{oleoyl-CoA} + \text{AMP} + PP_i$$
$$2PP_i + 2H_2O \longrightarrow 2P_i$$
$$\text{Glycerol 3-phosphate} + \text{palmitoyl-CoA} + \text{oleoyl-CoA} \longrightarrow \text{L-phosphatidate} + 2\text{CoA}$$

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L-Phosphatidate + $H_2O \longrightarrow$ 1,2-diacylglycerol + P_i

ATP + choline $\longrightarrow$ ADP + phosphocholine

CTP + phosphocholine $\longrightarrow$ CDP-choline + PP_i

$PP_i + H_2O \longrightarrow 2P_i$

CDP-choline + 1,2-diacylglycerol $\longrightarrow$ CMP + phosphatidylcholine

The net reaction is

Dihydroxyacetone phosphate + NADH + palmitate + oleate + 3ATP + CTP + choline + $4H_2O \longrightarrow$ phosphatidylcholine + NAD^+ + 2AMP + ADP + CMP + $5P_i$ + H^+

This can also be viewed as the sum of two equations:

Dihydroxyacetone phosphate + palmitate + oleate + choline + NADH + $2H^+ \longrightarrow$ phosphatidylcholine + NAD^+ + $3H_2O$

$3ATP + CTP + 7H_2O \longrightarrow 2AMP + ADP + CMP + 5P_i + 3H^+$

Note that the net equation shows the production of three nucleoside monophosphates (2 AMP + CMP). Recycling of these monophosphates to nucleoside diphosphates requires the input of 3 ATP.

$$2AMP + 2ATP \longrightarrow 4ADP$$
$$CMP + ATP \longrightarrow CDP + ADP$$

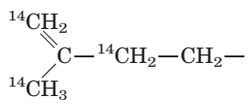
Furthermore, if we consider that CTP is energetically equivalent to ATP, then the synthesis of one phosphatidylcholine requires a total of 7 ATP.

12. Salvage Pathway for Synthesis of Phosphatidylcholine A young rat maintained on a diet deficient in methionine fails to thrive unless choline is included in the diet. Explain.

Answer The rat has two pathways for synthesizing phosphatidylcholine: the de novo pathway and the salvage pathway. The de novo pathway requires the transfer of a methyl group from *S*-adenosylmethionine (adoMet; see Fig. 21–27) to phosphatidylethanolamine. When the diet is deficient in methionine (an essential amino acid), the biosynthesis of adoMet and phosphatidylcholine is severely impaired. The salvage pathway, on the other hand, does not require adoMet but utilizes available choline. Thus, phosphatidylcholine can be synthesized when the diet is deficient in methionine, as long as choline is available.

13. Synthesis of Isopentenyl Pyrophosphate If 2-[^{14}C]acetyl-CoA is added to a rat liver homogenate that is synthesizing cholesterol, where will the ^{14}C label appear in Δ^3-isopentenyl pyrophosphate, the activated form of an isoprene unit?

Answer Three acetyl units are required for the synthesis of an isoprene unit (see Fig. 21–33). The ^{14}C label at C-2 of acetyl-CoA ends up in three places in the activated isoprene unit:



14. Activated Donors in Lipid Synthesis In the biosynthesis of complex lipids, components are assembled by transfer of the appropriate group from an activated donor. For example, the activated donor of acetyl groups is acetyl-CoA. For each of the following groups, give the form of the activated donor: **(a)** phosphate; **(b)** D-glucosyl; **(c)** phosphoethanolamine; **(d)** D-galactosyl; **(e)** fatty acyl; **(f)** methyl; **(g)** the two-carbon group in fatty acid biosynthesis; **(h)** Δ^3-isopentenyl.

Answer **(a)** ATP; **(b)** UDP-glucose; **(c)** CDP-ethanolamine; **(d)** UDP-galactose; **(e)** fatty acyl–CoA; **(f)** *S*-adenosylmethionine; **(g)** malonyl-CoA; **(h)** Δ^3-isopentenyl pyrophosphate.

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15. Importance of Fats in the Diet When young rats are placed on a totally fat-free diet, they grow poorly, develop a scaly dermatitis, lose hair, and soon die—symptoms that can be prevented if linoleate or plant material is included in the diet. What makes linoleate an essential fatty acid? Why can plant material be substituted?

Answer Linoleate is an unsaturated fatty acid required in the synthesis of prostaglandins. Animals lack the enzymes required to introduce double bonds into fatty acids beyond the Δ^9 position and therefore are unable to transform oleate to linoleate. Accordingly, linoleate is an essential fatty acid in animals. Plants can transform oleate to linoleate, and thus they provide animals with the required linoleate (Fig. 21–12).

16. Regulation of Cholesterol Biosynthesis Cholesterol in humans can be obtained from the diet or synthesized de novo. An adult human on a low-cholesterol diet typically synthesizes 600 mg of cholesterol per day in the liver. If the amount of cholesterol in the diet is large, de novo synthesis of cholesterol is drastically reduced. How is this regulation brought about?

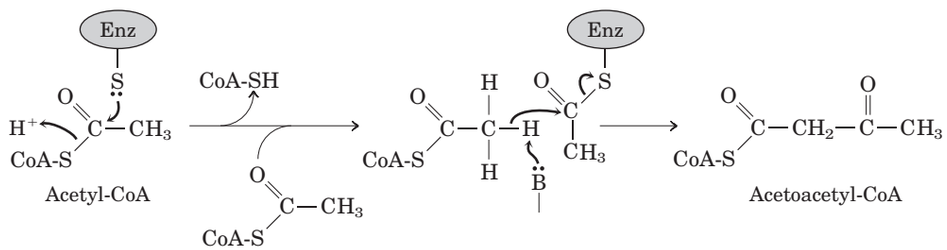
Answer The rate-determining step in the biosynthesis of cholesterol is the synthesis of mevalonate, catalyzed by HMG-CoA reductase. This enzyme is subject to feedback regulation by the negative modulator cholesterol. High intracellular [cholesterol] also reduces transcription of the gene that encodes HMG-CoA reductase.

17. Lowering Serum Cholesterol Levels with Statins Patients treated with a statin drug generally exhibit a dramatic lowering of serum cholesterol. However, the amount of the enzyme HMG-CoA reductase present in cells can increase substantially. Suggest an explanation for this effect.

Answer When cholesterol levels decline because of treatment with a statin, cells attempt to compensate by increasing expression of the gene encoding HMG-CoA reductase. The statins are good competitive inhibitors of HMG-CoA reductase activity nonetheless and reduce overall production of cholesterol.

18. Roles of Thiol Esters in Cholesterol Biosynthesis Draw a mechanism for each of the three reactions shown in Figure 21–34, detailing the pathway for the synthesis of mevalonate from acetyl-CoA.

Answer Note: There are several plausible alternatives that a student might propose in the absence of a detailed knowledge of the literature on this enzyme. *Thiolase reaction:* begins with nucleophilic attack of an active-site Cys residue on the first acetyl-CoA substrate, displacing —S-CoA and forming a covalent thioester link between Cys and the acetyl group. A base on the enzyme then extracts a proton from the methyl group of the second acetyl-CoA, leaving a carbanion that attacks the carbonyl carbon of the thioester formed in the first step. The sulfhydryl of the Cys residue is displaced, creating the product acetoacetyl-CoA.

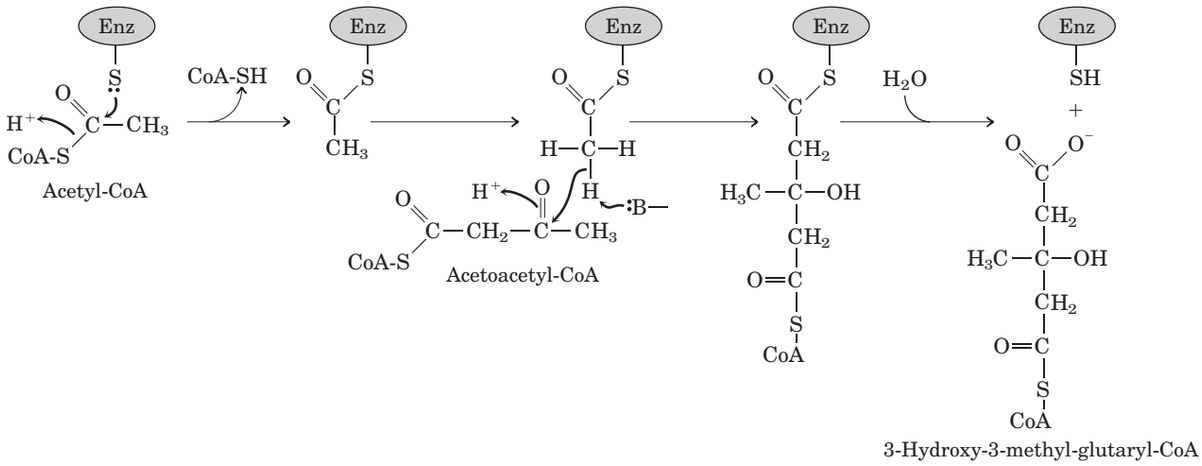


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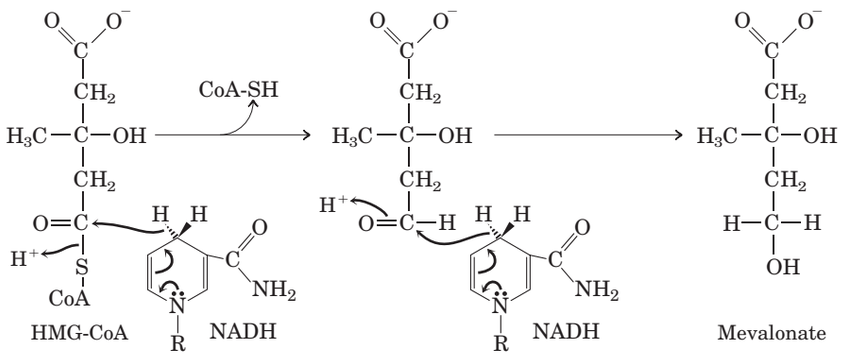
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HMG-CoA synthase reaction: begins in the same way, with a covalent thioester link formed between the enzyme's Cys residue and the acetyl group of acetyl-CoA, with displacement of the —S-CoA. The —S-CoA dissociates as CoA-SH, and acetoacetyl-CoA binds to the enzyme. A proton is abstracted from the methyl group of the enzyme-linked acetyl, forming a carbanion that attacks the ketone carbonyl of the acetoacetyl-CoA substrate. The carbonyl is converted to a hydroxyl ion in this reaction, and this is protonated to create —OH. The thioester link with the enzyme is then cleaved hydrolytically to generate the HMG-CoA product.



HMG-CoA reductase reaction: two successive hydride ions derived from NADPH first displace the —S-CoA, and then reduce the aldehyde to a hydroxyl group.



19. Potential Side Effects of Treatment with Statins Although clinical trials have not yet been carried out to document benefits or side effects, some physicians have suggested that patients being treated with statins also take a supplement of coenzyme Q. Suggest a rationale for this recommendation.

Answer Statins inhibit HMG-CoA reductase, an enzyme in the pathway to the synthesis of activated isoprenes, which are precursors of cholesterol and a wide range of isoprenoids, including coenzyme Q (ubiquinone). Hence, statins might reduce the levels of coenzyme Q available for mitochondrial respiration. Ubiquinone is obtained in the diet as well as by direct biosynthesis, but it is not yet clear how much is required and how well dietary sources can substitute for reduced synthesis. Reductions in the levels of particular isoprenoids may account for some side effects of statins.

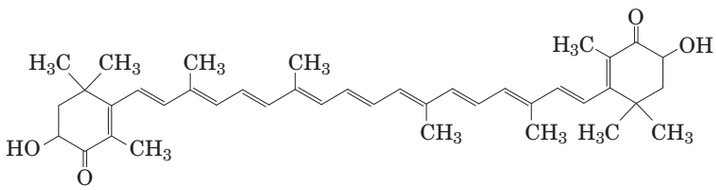
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Data Analysis Problem

20. Engineering *E. coli* to Produce Large Quantities of an Isoprenoid There is a huge variety of naturally occurring isoprenoids, some of which are medically or commercially important and produced industrially. The production methods include in vitro enzymatic synthesis, which is an expensive and low-yield process. In 1999, Wang, Oh, and Liao reported their experiments to engineer the easily grown bacterium *E. coli* to produce large amounts of astaxanthin, a commercially important isoprenoid.

Astaxanthin is a red-orange carotenoid pigment (an antioxidant) produced by marine algae. Marine animals such as shrimp, lobster, and some fish that feed on the algae get their orange color from the ingested astaxanthin. Astaxanthin is composed of eight isoprene units; its molecular formula is $C_{40}H_{52}O_4$:



Astaxanthin

(a) Circle the eight isoprene units in the astaxanthin molecule. Hint: Use the projecting methyl groups as a guide.

Astaxanthin is synthesized by the pathway shown on the following page, starting with Δ^3-isopentenyl pyrophosphate (IPP). Steps ① and ② are shown in Figure 21–36, and the reaction catalyzed by IPP isomerase is shown in Figure 21–35.

(b) In step ④ of the pathway, two molecules of geranylgeranyl pyrophosphate are linked to form phytoene. Is this a head-to-head or a head-to-tail joining? (See Figure 21–36 for details.)

(c) Briefly describe the chemical transformation in step ⑤.

(d) The synthesis of cholesterol (Fig. 21–37) includes a cyclization (ring closure) that involves a net oxidation by O_2. Does the cyclization in step ⑥ of the astaxanthin synthetic pathway require a net oxidation of the substrate (lycopene)? Explain your reasoning.

E. coli does not make large quantities of many isoprenoids, and does not synthesize astaxanthin. It is known to synthesize small amounts of IPP, DMAPP, geranyl pyrophosphate, farnesyl pyrophosphate, and geranylgeranyl pyrophosphate. Wang and colleagues cloned several of the *E. coli* genes that encode enzymes needed for astaxanthin synthesis in plasmids that allow their overexpression. These genes included *idi,* which encodes IPP isomerase, and *ispA,* which encodes a prenyl transferase that catalyzes steps ① and ②.

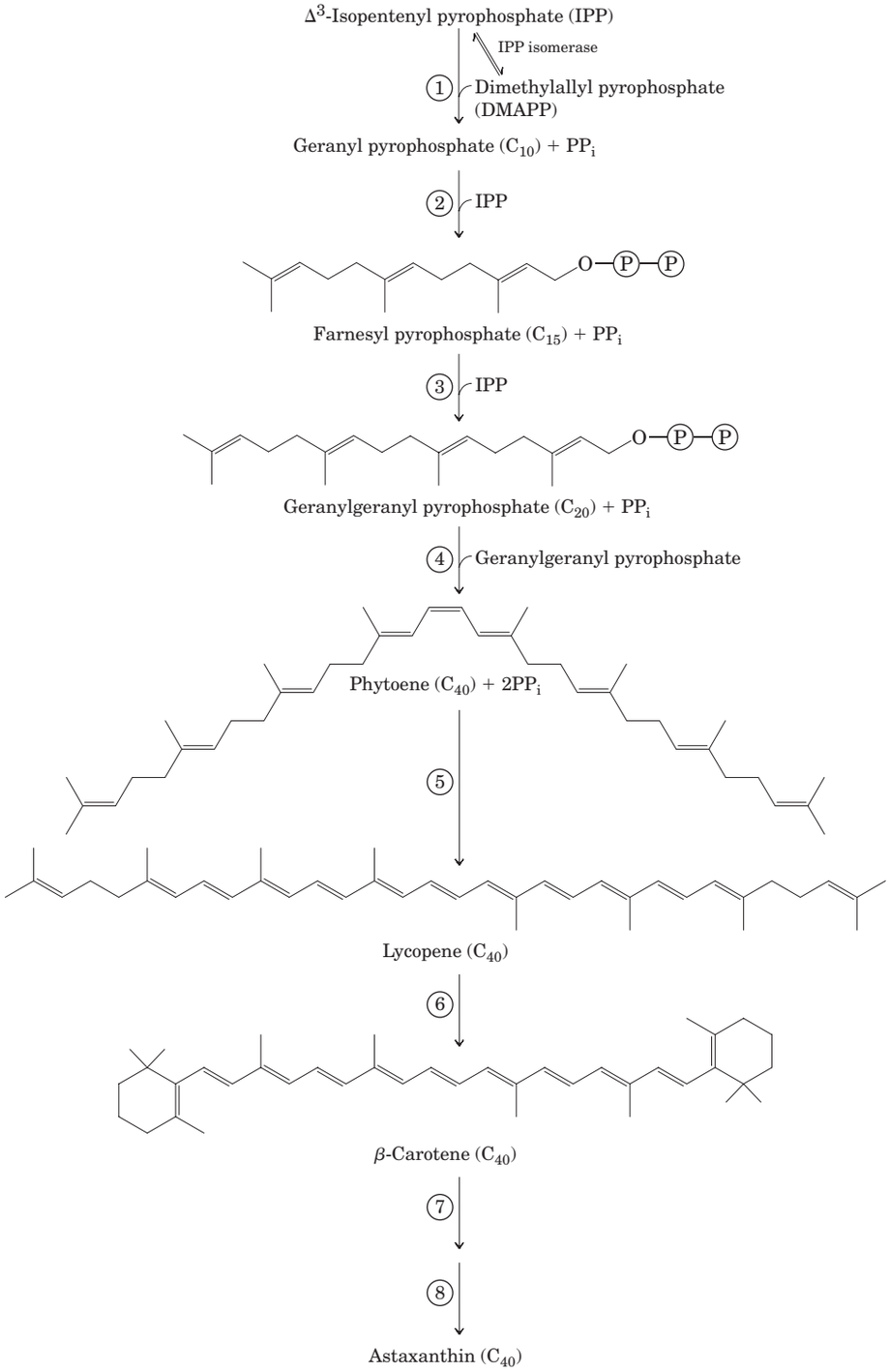
To engineer an *E. coli* capable of the complete astaxanthin pathway, Wang and colleagues cloned several genes from other bacteria into plasmids that would allow their overexpression in *E. coli.* These genes included *crtE* from *Erwinia uredovora,* which encodes an enzyme that catalyzes step ③; and *crtB, crtI, crtY, crtZ,* and *crtW* from *Agrobacterium aurantiacum,* which encode enzymes for steps ④, ⑤, ⑥, ⑦, and ⑧, respectively.

The investigators also cloned the gene *gps* from *Archaeoglobus fulgidus*, overexpressed this gene in *E. coli*, and extracted the gene product. When this extract was reacted with [^{14}C]IPP and DMAPP, or geranyl pyrophosphate, or farnesyl pyrophosphate, only ^{14}C-labeled geranylgeranyl pyrophosphate was produced in all cases.

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(e) Based on these data, which step(s) in the pathway are catalyzed by the enzyme encoded by *gps*? Explain your reasoning.

Wang and coworkers then constructed several *E. coli* strains overexpressing different genes and measured the orange color of the colonies (wild-type *E. coli* colonies are off-white) and the amount of astaxanthin produced. Their results are shown below.

Strain	Gene(s) overexpressed	Orange color	Astaxanthin yield (μg/g dry weight)
1	*crtBIZYW*	−	ND
2	*crtBIZYW, ispA*	−	ND
3	*crtBIZYW, idi*	−	ND
4	*crtBIZYW, idi, ispA*	−	ND
5	*crtBIZYW, crtE*	+	32.8
6	*crtBIZYW, crtE, ispA*	+	35.3
7	*crtBIZYW, crtE, idi*	++	234.1
8	*crtBIZYW, crtE, idi, ispA*	+++	390.3
9	*crtBIZYW, gps*	+	35.6
10	*crtBIZYW, gps, idi*	+++	1,418.8

Note: ND, not determined

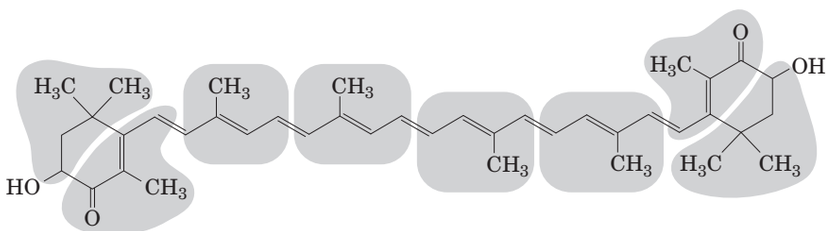
(f) Comparing the results for strains 1 through 4 with those for strains 5 through 8, what can you conclude about the expression level of an enzyme capable of catalyzing step ③ of the astaxanthin synthetic pathway in wild-type *E. coli?* Explain your reasoning.

(g) Based on the data above, which enzyme is rate-limiting in this pathway, IPP isomerase or the enzyme encoded by *idi*? Explain your reasoning.

(h) Would you expect a strain overexpressing *crtBIZYW*, *gps*, and *crtE* to produce low (+), medium (++), or high (+++) levels of astaxanthin, as measured by its orange color? Explain your reasoning.

Answer

(a)



(b) Head-to-head. There are two ways to look at this. First, the "tail" of geranylgeranyl pyrophosphate has a branched dimethyl structure, as do both ends of phytoene. Second, no free —OH is formed by the release of PP_i, indicating that the two —O—Ⓟ—Ⓟ "heads" are linked to form phytoene.

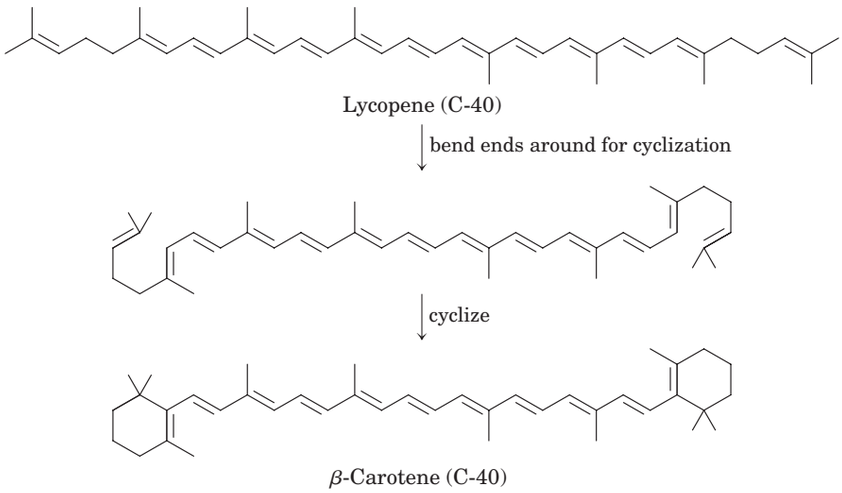
(c) Four rounds of dehydrogenation convert four single bonds to double bonds.

(d) No. A count of single and double bonds in the following reaction shows that one double bond is replaced by two single bonds—so, there is no net oxidation or reduction.

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(e) Steps ① through ③. The enzyme can convert IPP and DMAP to geranylgeranyl pyrophosphate, but catalyzes no further reactions in the pathway, as confirmed by results with the other substrates.

(f) Strains 1 through 4 lack *crtE* and have much lower astaxanthin production than strains 5 through 8, all of which overexpress *crtE*. Thus, overexpression of *crtE* leads to a substantial increase in astaxanthin production. Wild-type *E. coli* has some step ③ activity, but this conversion of farnesyl pyrophosphate to geranylgeranyl pyrophosphate is strongly rate-limiting.

(g) IPP isomerase. Comparing strains 5 and 6 shows that adding *ispA*, which catalyzes steps ① and ②, has little effect on astaxanthin production, so these steps are not rate-limiting. However, comparing strains 5 and 7 shows that adding *idi* substantially increases astaxanthin production, so IPP isomerase must be the rate-limiting step when *crtE* is overexpressed.

(h) A low (+) level, comparable to that of strains 5, 6, and 9. Without overexpression of *idi*, production of astaxanthin is limited by low IPP isomerase activity and the resulting limited supply of IPP.

Reference

Wang, C.-W., Oh, M.-K., & Liao, J.C. (1999) Engineered isoprenoid pathway enhances astaxanthin production in *Escherichia coli*. *Biotechnol. Bioeng.* **62,** 235–241.

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